

Generative Modeling via SDEs

From DDPM & SMLD to a Continuous-Time View

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

Overview

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Unifying SMLD & DDPM

The corrupt-then-reverse paradigm — both are score-based models.

2

Why an SDE Framework

Flexible sampling, better samplers & likelihoods, controllable generation.

3

Forward & Reverse SDEs

The core objects: diffusing data into noise and reversing with the score.

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Deriving SDEs from Discrete Limits

VP from DDPM, VE from the SMLD noise ladder.

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The VP SDE

Variance-preserving diffusion — the continuous analog of DDPM.

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The VE SDE

Variance-exploding diffusion — the continuous analog of SMLD.

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Probability Flow ODE & the Unified Landscape

Deterministic sampling, exact likelihoods, one model many solvers.

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Experimental Results

CIFAR-10 — sampler comparison, exact likelihoods, and SOTA sample quality.

Central object: the score $\nabla_x \log p_t(x)$ ties every method together. · Appendix: full VP α/σ derivation.

PART 01

Unifying SMLD & DDPM

The corrupt-then-reverse paradigm

SMLD · DDPM · The unifying insight · The score function

The "Corrupt Then Reverse" Paradigm

The unifying intuition behind SMLD, DDPM, and the SDE framework.

1 · SEQUENTIALLY CORRUPT DATA

Slowly inject noise to transform the complex data distribution into a tractable prior (e.g., a Gaussian).

2 · LEARN TO REVERSE THE CORRUPTION

Train a model to undo the noise step-by-step, transforming the prior back into new data samples.

Corrupt forward into noise → learn the reverse → generate.

SMLD: Score Matching with Langevin Dynamics

TRAINING PHASE

Perturb data with a discrete ladder of noise scales:

$$\sigma_1 < \sigma_2 < \dots < \sigma_k$$

Estimate the score at each scale via denoising score matching:

$$s\theta(x, \sigma) \approx \nabla_x \log p\sigma(x)$$

GENERATION PHASE

Run Annealed Langevin Dynamics.

Step through decreasing noise scales from noise to data:

$$\sigma_k \rightarrow \sigma_{k-1} \rightarrow \dots \rightarrow \sigma_1$$

At each scale, take Langevin MCMC steps guided by the estimated score toward high-density regions.

DDPM: Denoising Diffusion Probabilistic Models

TRAINING PHASE

Define a fixed Markov chain that gradually adds Gaussian noise over K steps.

Train a sequence of probabilistic reverse models step-by-step:

$$p_{\theta}(x_{t-1} | x_t)$$

Tractable because the reverse conditionals have a known Gaussian form.

GENERATION PHASE

Start from pure Gaussian noise:

$$x_k \sim N(0, I)$$

Iteratively denoise the sample:

$$x_k \rightarrow x_{k-1} \rightarrow \dots \rightarrow x_0$$

Each step removes a small amount of noise using the learned reverse model.

Unifying Viewpoint: Both Are Score-Based

THE UNIFYING INSIGHT

In continuous state spaces, DDPM training implicitly computes scores at each noise scale.

WHY THEY ARE DEEPLY RELATED

- ✓ Both estimate scores (gradients of log-density) of intermediate noisy distributions.
- ✓ Both use those estimated scores to guide a reverse denoising process.
- ✓ Apparent differences (Markov chain vs. Langevin dynamics) are merely implementation details of the same underlying principle.

The Core Quantity: Scores Along the Corruption Path

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

The score of the marginal distribution $p_t(x)$ at time t

DIRECTIONAL GUIDE

Points toward regions of higher data density
— the exact direction needed to denoise.

NORMALIZATION-FREE

Estimated without knowing the intractable
normalizing constant of $p_t(x)$.

TRACTABLE TRAINING

Denoising score matching is scalable and
computationally efficient.

SMLD and DDPM are unified as score-based models — differing only in how they parameterize the noise process.

PART 02

Why an SDE Framework?

Enabling new samplers and capabilities

Flexible sampling · Better samplers & likelihoods · Controllable generation

Why an SDE Framework?

PRIMARY GOAL

Extend score-based models by viewing them through stochastic differential equations.

01 · FLEXIBLE SAMPLING

Any general-purpose SDE solver can integrate the reverse process — expanding the design space beyond fixed discrete samplers.

02 · BETTER SAMPLERS & LIKELIHOODS

Unlocks Predictor–Corrector methods and the Probability Flow ODE, enabling exact likelihood computation.

03 · CONTROLLABLE GENERATION

Condition on information not available during training (e.g., inpainting) without retraining the model.

Motivation (A): Flexible Sampling

Once posed as an SDE, the reverse process can be integrated by any general-purpose solver.

THE SOLVER TOOLBOX

- Euler–Maruyama
- Milstein method
- Stochastic Runge–Kutta
- Predictor–Corrector methods

EXPANDED CAPABILITIES

This expands the design space beyond fixed K-step discrete samplers.

Trade off speed vs. accuracy using adaptive step sizes and error-control mechanisms.

Discrete DDPM and SMLD samplers become just two specific choices within a much larger space.

Motivation (B): Better Samplers & Exact Likelihoods

PREDICTOR-CORRECTOR (PC)

Combines a numerical SDE solver step (predictor) with a score-based MCMC step (corrector).

Benefit:

Improved sample quality at the same computational budget.

PROBABILITY FLOW ODE

A deterministic ODE that shares the same marginal distributions as the stochastic SDE.

Benefit:

Fast adaptive sampling using black-box ODE solvers.

The Big Payoff — Exact Likelihood: the ODE view computes the exact likelihood of data via the instantaneous change-of-variables formula — not directly available in the stochastic formulation.

Motivation (C): Controllable Generation

Modulate generation by conditioning on new information — without retraining the model.

THE MECHANISM: BAYES' RULE ON SCORES

$$\nabla_x \log p_t(x \mid y) = \nabla_x \log p_t(x) + \nabla_x \log p_t(y \mid x)$$

Conditional score = unconditional score (learned) + guidance term (estimated)

APPLICATIONS — FROM ONE UNCONDITIONAL MODEL

- Image inpainting
- Colorization
- Class-conditioning
- Inverse problems (super-resolution)

From Discrete Noise Ladders to a Continuum

THE DISCRETE APPROACH

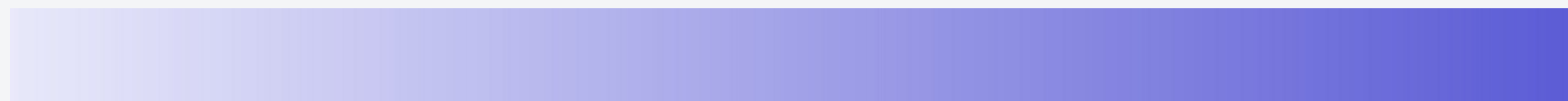
$(\sigma_1, \dots, \sigma_k)$ or $(\beta_1, \dots, \beta_k)$

SMLD and DDPM rely on a finite set of noise levels or steps.

THE CONTINUOUS VIEW

$\{p_t\}$ for $t \in [0, 1]$

Consider a continuum of distributions indexed by continuous time t .



$t = 0$ · Data distribution p_{data}

$t = 1$ · Prior $N(0, I)$

PART 03

Forward & Reverse SDEs

The core objects of the framework

Forward SDE · Reverse-time SDE · PC samplers · Probability Flow ODE

Core Idea: The Forward SDE

Diffusing data into noise.

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w}$$

$f(\mathbf{x}, t)$: drift coefficient · $g(t)$: diffusion coefficient · $d\mathbf{w}$: Brownian motion

FIXED PROCESS

Contains no trainable parameters. Prescribed by design to match a specific noise schedule.

DATA-INDEPENDENT

The process does not depend on the data distribution; it blindly adds noise.

Core Idea: The Reverse-Time SDE

For a forward SDE of the form on slide 16, the reverse is itself an SDE (Anderson, 1982).

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g(t) d\bar{\mathbf{w}}$$

Same f and g as the forward SDE (slide 16) — to reverse it, we only need the score of the marginals.

1 Train Score Model

Train a time-dependent network via score matching: $s_{\theta}(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

2 Substitute

Plug the learned score s_{θ} into the reverse-time SDE in place of the true score.

3 Integrate

Solve the SDE numerically from $t=1$ (noise) to $t=0$ (data) to generate samples.

Derivation: the reverse-time SDE is proved in Anderson (1982); see also Song et al. (2021), Appendix A.

Contribution #1: A Sampling Toolbox (PC Samplers)

THE PREDICTOR

Numerical SDE step.

Uses a standard solver (e.g., Euler–Maruyama) to estimate the sample at the next time step.

THE CORRECTOR

Score-based MCMC.

Runs Langevin dynamics or HMC to correct the marginal distribution and refine the sample.

THE RESULT

Predictor + Corrector $\Rightarrow$ higher sample quality at the same compute budget.

Contribution #2: The Probability Flow ODE

For the same forward SDE on slide 16: a deterministic ODE with the same marginals as the reverse SDE.

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

Same f and g as the forward SDE (slide 16); note the factor of $\frac{1}{2}$ on the score vs. the reverse SDE.

FAST ADAPTIVE SAMPLING

Use black-box ODE solvers (e.g., Dormand–Prince) with adaptive step sizes.

FLEXIBLE LATENT SPACE

Deterministic encoding of data to noise — interpolation and manipulation in latent space.

EXACT LIKELIHOODS

Via the instantaneous change-of-variables formula, avoiding ELBO approximations.

Derivation: the probability-flow ODE is derived in Song et al. (2021), Appendix D.1, from the same forward SDE.

Why It Works: Stay on the Forward Marginals

The network gives the score at a point — reliable only where the distribution is correct.



IT KNOWS THE SCORE, NOT THE DISTRIBUTION

At a point x and time t the network returns the score — the local direction toward higher density. It was trained on samples from p_t , so it is only accurate where p_t has mass.

CORRECT IN → CORRECT OUT

If the population at time t is distributed as p_t , stepping every point along the score makes the next population correct too. Start from the right prior and correctness propagates down to the data.

OFF-DISTRIBUTION → IT BREAKS

If points drift off p_t , the network is queried out-of-distribution and returns wrong scores, so errors compound. The one thing that matters: keep each intermediate marginal equal to the forward p_t .

At each time, match the reverse process's p_t to the forward process's p_t , so the network produces an accurate score.
Hence the probability-flow ODE (previous slide) — which maintains exactly these marginals — yields valid deterministic sampling.

Contribution #3: A Unified Framework

SMLD and DDPM as discretizations of two continuous-time SDEs.

VE SDE · VARIANCE EXPLODING

SMLD → the variance of the process grows without bound as $t \rightarrow \infty$.

VP SDE · VARIANCE PRESERVING

DDPM → the variance is bounded (typically at 1) throughout the process.

Key implication: the familiar discrete algorithms are not fundamentally different — they are two specific discretizations of two continuous-time SDEs within the same framework.

Roadmap: VP vs. VE and Next Steps

1 Choose Forward SDE

Select a noise process (e.g., VE, VP) to transform data into the prior.

2 Learn Score

Train $s\theta(x,t) \approx \nabla \log p_t(x)$ via score matching.

3 Sample

Integrate the reverse SDE or the probability-flow ODE.

VP SDE · VARIANCE PRESERVING

A forward-SDE diffusion (noising) process.

Shrinks the signal as it adds noise, so the total variance stays bounded (≈ 1).

→ **The continuous analog of DDPM.**

VE SDE · VARIANCE EXPLODING

A forward-SDE diffusion (noising) process.

Keeps adding ever-larger noise with no signal decay, so the variance grows without bound.

→ **The continuous analog of SMLD.**

VP and VE are the two canonical SDE diffusion methods — forward processes that turn data into noise, differing only in how the variance behaves.

PART 04

Deriving SDEs from Discrete Limits

VP from DDPM · VE from the SMLD ladder

Taylor expansion · Brownian increments · Forward & reverse SDEs

Goal: Derive Forward SDEs from Discrete Processes

DDPM FORWARD CHAIN → VP SDE

The variance-preserving limit of the discrete Gaussian chain.

SMLD NOISE LADDER → VE SDE

The variance-exploding limit of the discrete noise ladder.

THE STRATEGY

- 1 Rescale noise to make each step infinitesimal.
- 2 Write the increment $x(t+\Delta t) - x(t)$.
- 3 Take the limit $\Delta t \rightarrow 0$.
- 4 Identify the drift and diffusion terms.

VP: Start from the DDPM Forward Chain

$$\mathbf{x}_i = \sqrt{1 - \beta_i} \mathbf{x}_{i-1} + \sqrt{\beta_i} \mathbf{z}_{i-1}$$

where $\mathbf{z}_{i-1} \sim N(0, I)$

SCALING FOR $N \rightarrow \infty$

Define $\bar{\beta}_i = N \cdot \beta_i$.

This keeps the total noise variance over $[0,1]$ finite as the number of steps increases.

TIME VARIABLES

Let $t = i/N$ and $\Delta t = 1/N$.

Thus $\beta_i = \bar{\beta}(t) \cdot \Delta t$.

Rewrite in Continuous-Time Form

$$\mathbf{x}(t + \Delta t) = \sqrt{1 - \beta(t + \Delta t)\Delta t} \mathbf{x}(t) + \sqrt{\beta(t + \Delta t)\Delta t} \mathbf{z}(t)$$

Now we expand this for small Δt ...

Taylor Expand the Drift Term

APPROXIMATION

Approximation: $\sqrt{1 - \varepsilon} \approx 1 - \frac{1}{2}\varepsilon$

Apply to drift: $\sqrt{1 - \beta(t)\Delta t} \approx 1 - \frac{1}{2}\beta(t)\Delta t$

Substitute back: $\mathbf{x}(t+\Delta t) \approx (1 - \frac{1}{2}\beta(t)\Delta t) \cdot \mathbf{x}(t) + \sqrt{\beta(t)\Delta t} \cdot \mathbf{z}(t)$

$$\mathbf{x}(t + \Delta t) - \mathbf{x}(t) \approx -\frac{1}{2}\beta(t)\mathbf{x}(t)\Delta t + \sqrt{\beta(t)\Delta t} \mathbf{z}(t)$$

Identify the Brownian Increment

$$\sqrt{\Delta t} \mathbf{z}(t) \equiv d\mathbf{w}_t \quad (\text{since } \mathbf{z}(t) \sim N(0, I))$$

RESULTING VP FORWARD SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}_t$$

Taking the limit $\Delta t \rightarrow 0$ yields this linear SDE.

VE: Start from Incremental Variance

SMLD GOAL

$$\mathbf{x}_k = \mathbf{x}_0 + \sigma_k \boldsymbol{\varepsilon}$$

MARKOV CHAIN CONSTRUCTION

To satisfy the marginals, the incremental step must be:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \sqrt{\sigma_{k+1}^2 - \sigma_k^2} \mathbf{z}_k$$

$$\mathbf{z}_k \sim N(0, I)$$

Take the Continuous-Time Limit for VE

DERIVATION

- 1 · **Variance difference:** $\sigma^2(t+\Delta t) - \sigma^2(t) \approx d/dt[\sigma^2(t)] \cdot \Delta t$
- 2 · **Substitute into increment:** $x(t+\Delta t) - x(t) = \sqrt{(d/dt[\sigma^2(t)] \cdot \Delta t)} \cdot z(t)$
- 3 · **Identify Brownian motion:** $\sqrt{\Delta t} \cdot z(t) = dw_t$

RESULTING VE FORWARD SDE

$$dx = \sqrt{\frac{d}{dt}\sigma^2(t)} dw$$

Summary of Forward SDEs

VP (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}_t$$

VE (SMLD)

$$d\mathbf{x} = \sqrt{\frac{d}{dt}\sigma^2(t)} d\mathbf{w}_t$$

Both are linear Itô diffusions. The key difference is the presence of the drift term in VP.

Reverse-Time SDE (Anderson, 1982)

FORWARD

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w}_t$$

REVERSE

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - g^2(t) \nabla \log p_t(\mathbf{x}) \right] dt + g(t) d\bar{\mathbf{w}}_t$$

Reversal introduces an additional drift term, $-g^2(t) \cdot \text{score}$. This is why estimating the score lets us reverse the diffusion.

Reverse SDE for VP

$$f(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x} \quad g(t) = \sqrt{\beta(t)}$$

RESULTING REVERSE VP SDE

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t)s_{\theta}(\mathbf{x}, t) \right] dt + \sqrt{\beta(t)} d\bar{\mathbf{w}}$$

Replace the unknown score $\nabla \log p_t(\mathbf{x})$ with the trained network $s_{\theta}(\mathbf{x}, t)$.

Reverse SDE for VE

$$f(\mathbf{x}, t) = 0 \quad g(t) = \sqrt{d/dt[\sigma^2(t)]}$$

RESULTING REVERSE VE SDE

$$d\mathbf{x} = \left[-g^2(t) s_{\theta}(\mathbf{x}, t) \right] dt + g(t) d\bar{\mathbf{w}}$$

Notice: no anchor drift term ($-\frac{1}{2}\beta\mathbf{x}$) here. The reverse process is driven purely by the score.

What We Achieved

FORWARD SDES

- VP derived from the DDPM limit
- VE derived from the noise-ladder limit

REVERSE SDES

- Obtained directly via Anderson's theorem
- No posterior derivations required
- No discrete reverse conditionals needed

Sampling now reduces to a standard numerical problem: solving the reverse SDE.

PART 05

The VP SDE

Variance-preserving diffusion — the continuous analog of DDPM

Bounded variance · Closed-form marginals · Reverse SDE · DDPM samplers

VP SDE: Variance-Preserving Diffusion

CORE DEFINITION

Variance stays bounded as time increases, preventing the signal scale from blowing up.

THE MECHANISM

Slowly attenuate the signal while injecting noise, so the total variance stays controlled (often ≈ 1).

THE CONNECTION

The direct continuous-time analog of DDPM-style discrete noising schedules.

VP Forward SDE Definition

[Derived → slide 28](#)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}_t$$

DRIFT TERM · $-\frac{1}{2}\beta(t)\mathbf{x}$

Pulls the data toward 0, attenuating the signal to keep variance controlled.

DIFFUSION TERM · $\sqrt{\beta(t)} d\mathbf{w}_t$

Injects noise at a rate set by the nonnegative schedule $\beta(t)$.

VP: From SDE to Closed-Form Solution

[Full derivation →](#)

The VP SDE is linear, so its transition kernel $p_t(x | x_0)$ is Gaussian — fixed entirely by a mean and a variance.

THE MEAN · SIGNAL DECAY

Drop the noise: the drift $-\frac{1}{2}\beta(t)x$ only rescales the signal.

Its integrating factor gives the signal's decay $\alpha(t)$:

$$\alpha(t) = \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right)$$

THE VARIANCE · NOISE BUILD-UP

Injected noise builds up as $d\Sigma/dt = -\beta(t)\Sigma + \beta(t)$, from $\Sigma(0)=0$.

Solving this linear ODE gives $\sigma^2(t)$:

$$\sigma^2(t) = 1 - \alpha^2(t)$$

THE SAME PROCESS, IN CLOSED FORM

$$\mathbf{x}(t) = \alpha(t) \mathbf{x}(0) + \sigma(t) \mathbf{z}, \quad \mathbf{z} \sim N(0, I)$$

Exactly the solution of the VP forward SDE — the differential equation and this closed form describe the same process, so you can jump to any time t in one step.

VP: Closed-Form Marginals

$$\mathbf{x}(t) = \alpha(t)\mathbf{x}(0) + \sigma(t)\mathbf{z}, \quad \mathbf{z} \sim N(0, I)$$

$$\alpha(t) = \exp(-\frac{1}{2}\int\beta(s)ds) \quad \sigma^2(t) = 1 - \alpha^2(t)$$

Key property — Bounded Variance: because $\alpha^2(t) + \sigma^2(t) = 1$, the total variance of $\mathbf{x}(t)$ stays controlled (typically ≈ 1) throughout the process, preventing numerical instability.

VP: Score-Model Target

$$s_{\theta}(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

DIRECT SCORE

Train the network to output the score vector directly.

NOISE PREDICTION

Predict the noise ϵ_{θ} and convert:

$$\mathbf{s}_{\theta} = -\epsilon_{\theta} / \sigma(t)$$

DDPM's noise-prediction is essentially a reparameterization of the score in this Gaussian setting.

VP: Reverse-Time SDE

[Derived → slide 33](#)

GENERAL REVERSE TEMPLATE

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - g^2(t) \nabla \log p_t(\mathbf{x}) \right] dt + g(t) d\bar{\mathbf{w}}$$

For VP: $f(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}$ $g(t) = \sqrt{\beta(t)}$

RESULTING REVERSE VP SDE

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t)s_\theta(\mathbf{x}, t) \right] dt + \sqrt{\beta(t)} d\bar{\mathbf{w}}$$

VP: Discretizations & DDPM

Discretize the VP reverse process with a specific timestep rule, and you recover DDPM-like samplers.

ANCESTRAL SAMPLING

Corresponds to using the reverse conditional Gaussian $p(x_{t-1}|x_t)$ at each step — the standard DDPM sampler.

SOLVER FLEXIBILITY

Different discretizations (Euler–Maruyama, higher-order methods) yield different algorithms from the same model.

VP: Sampling in Practice

SCHEDULE SELECTION

Performance depends heavily on a good $\beta(t)$ schedule. Cosine schedules often outperform linear ones by preserving signal longer.

DISCRETIZATION CHOICE

Deterministic variants like DDIM are possible via the ODE view, offering faster sampling.

Empirically, VP/DDPM gives strong perceptual quality with standard architectures and samplers.

VP SDE: Summary



Forward Process

Bounded variance, linear drift to 0, plus diffusion.



Training

Estimate the time-dependent score (or noise).



Sampling

Integrate the reverse SDE (stochastic) or the probability-flow ODE (deterministic).



Connection

Discretizations connect directly to the DDPM family.

PART 06

The VE SDE

Variance-exploding diffusion — the continuous analog of SMLD

Exploding variance · DSM · Reverse SDE · Predictor–Corrector

VE SDE: Variance-Exploding Diffusion

CORE DEFINITION

Noise scale grows dramatically with time — the variance 'explodes'.

THE MECHANISM

Instead of damping the signal, simply add more and more noise until the distribution is effectively pure Gaussian.

THE CONNECTION

Matches the noise-level ladder intuition in SMLD (Song & Ermon, 2019).

VE Forward SDE Definition

[Derived → slide 30](#)

$$d\mathbf{x} = g(t) d\mathbf{w}_t$$

PURE DIFFUSION

No drift term. The process is driven purely by Brownian motion scaled by $g(t)$.

NOISE SCALE

$g(t)$ is chosen so the noise standard deviation increases from $\sigma_{\min}$ to $\sigma_{\max}$.

VE: Closed-Form Marginals

$$\mathbf{x}(t) = \mathbf{x}(0) + \sigma(t)\mathbf{z}, \quad \mathbf{z} \sim N(0, I)$$

Pure diffusion: data convolved with Gaussian noise

Connection to SMLD: this is exactly the object SMLD trains on — scores of Gaussian-smoothed densities across a range of noise levels $\sigma(t)$.

VE: Score Model & DSM

$$s_{\theta}(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

CONTINUUM OF SIGMAS

Corresponds to Denoising Score Matching (DSM) across a continuous range of noise scales.

SMLD CONNECTION

SMLD's 'discrete sigmas + Langevin per sigma' is simply a discretized version of this continuous-time story.

We learn the gradient of the log-density for every noise level in the explosion process.

VE: Reverse-Time SDE

[Derived → slide 34](#)

Forward parameters: $f(\mathbf{x}, t) = 0$ $g(t) = \text{chosen schedule}$

RESULTING REVERSE VE SDE

$$d\mathbf{x} = \left[-g^2(t) s_{\theta}(\mathbf{x}, t) \right] dt + g(t) d\bar{\mathbf{w}}$$

Compare to VP: no anchor-drift term ($-\frac{1}{2}\beta\mathbf{x}$) here. The drift is purely score-driven.

VE: Discretizations & SMLD

The SMLD sampler is essentially a specific Predictor–Corrector discretization of the VE SDE.

PREDICTOR STEP

A standard numerical SDE solver step (e.g., Euler–Maruyama) moves the sample backward in time.

CORRECTOR STEP

Langevin MCMC steps (at fixed time t) correct the sample to match the marginal $p_t(x)$.

Result: recovering SMLD-style annealed Langevin dynamics.

VE: Practical Sampling — Why PC Shines

PREDICTOR

Step forward — makes progress along time, solving the reverse SDE.

CORRECTOR (LANGEVIN)

Re-equilibrate — pulls the sample toward the marginal $p_t(x)$ at the current noise level.

Why it matters: with large noise scales the score landscape changes rapidly; the corrector stabilizes the trajectory by pulling samples back to the high-probability manifold.

VE SDE: Summary



Forward Process

Pure diffusion; noise std increases from $\sigma_{\min}$ $\rightarrow$ $\sigma_{\max}$.



Training

Scores of Gaussian-smoothed densities (DSM).



Sampling

Reverse SDE, often with Predictor–Corrector steps.



Connection

Discretizations connect directly to the SMLD family.

PART 07

Probability Flow ODE & the Unified Landscape

Deterministic sampling, exact likelihoods, one model — many solvers

Probability Flow ODE · Matching marginals · SDE vs ODE · The bridge

Probability Flow: Why?

With the same learned score $s_\theta(x,t)$, we can define deterministic dynamics.

DETERMINISM

Avoids sampling noise. Given an initial noise vector, the output is fixed.

ADAPTIVE SOLVERS

Enables black-box ODE solvers (RK45, etc.) for error control and speed.

EXACT LIKELIHOODS

A clean route to log-likelihoods via the change-of-variables formula.

Probability Flow ODE Definition

[Introduced → slide 19](#)

GIVEN FORWARD SDE

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w}_t$$

PROBABILITY FLOW ODE

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla \log p_t(\mathbf{x}) \right] dt$$

In practice, replace the true score $\nabla \log p_t(\mathbf{x})$ with the learned model $s_\theta(\mathbf{x}, t)$.

ODE Property: Matching Marginals

CRUCIAL GUARANTEE

The Probability Flow ODE shares the exact same marginal distributions $p_t(x)$ as the corresponding stochastic SDE.

THE IMPLICATION

Integrating this ODE from 'noise time' back to 'data time' yields samples from the same target distribution (in the idealized limit).

VP Probability Flow ODE

[General form → slide 57](#)

$$\mathbf{f} = -\frac{1}{2}\beta(t)\mathbf{x} \quad \mathbf{g}^2 = \beta(t)$$

VP PROBABILITY FLOW ODE

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \frac{1}{2}\beta(t)s_{\theta}(\mathbf{x}, t) \right] dt$$

Deterministic alternative to DDPM/VP reverse-SDE sampling. Note the $\frac{1}{2}$ on the score term.

VE Probability Flow ODE

[General form → slide 57](#)

$$f = 0 \quad g^2 = g^2(t)$$

VE PROBABILITY FLOW ODE

$$d\mathbf{x} = \left[-\frac{1}{2}g^2(t)s_{\theta}(\mathbf{x}, t) \right] dt$$

A deterministic flow guided purely by the score — simple, drift-free structure.

ODE Payoff: Adaptive Sampling

ERROR CONTROL

ODE solvers automatically adjust step sizes based on local error estimates.

DYNAMIC ALLOCATION

Spend more steps where dynamics are stiff (curved) and fewer where the flow is straight.

Number of Function Evaluations (NFE) becomes a runtime knob — not a fixed hyperparameter K .

ODE Payoff: Exact Likelihoods

$$\log p(\mathbf{x}(0)) = \log p(\mathbf{x}(T)) - \int_0^T \text{tr}(\nabla_{\mathbf{x}} f_{\text{ode}}) dt$$

CONTINUOUS NORMALIZING FLOW

The ODE is a continuous invertible mapping — compute log-likelihood via the instantaneous change-of-variables formula.

HUTCHINSON'S TRICK

The trace term is estimated stochastically to avoid the full Jacobian, scaling to high dimensions.

SDE vs. ODE: When to Use Which

REVERSE SDE

- ✓ Stochasticity can correct errors
- ✓ Robust to score inaccuracies
- ✓ Supports Predictor–Corrector sampling

PROBABILITY FLOW ODE

- ✓ Deterministic (fixed latent map)
- ✓ Adaptive solvers (faster)
- ✓ Exact likelihood computation
- ✓ Latent-space manipulation

Crucially, both methods use the exact same trained score model $s_\theta(x,t)$.

Bridge: The Unified Landscape

VP SDE DISCRETIZATIONS

→ DDPM family

VE SDE DISCRETIZATIONS

→ SMLD family

SDE framework: one score model + many solvers / discretizations.

Bridge: Dimensions of Comparison

SCHEDULES

$\beta(t)$ for VP
 $\sigma(t)$ / $g(t)$ for VE

SAMPLERS

Ancestral (DDPM)
DDIM (deterministic)
Euler–Maruyama
Predictor–Corrector
ODE (adaptive)

TRADE-OFFS

Sample quality
Speed (NFE)
Likelihood
Controllability

One framework, many knobs — quality, speed, likelihood, and control.

PART 08

Experimental Results

Song et al. (2021) — what the SDE framework delivers on CIFAR-10

Samplers · Likelihoods · Sample quality

Results at a Glance

CIFAR-10, unconditional generation — and the first high-fidelity score-based 1024^2 samples.

2.20

FID ↓

New SOTA on CIFAR-10 (NCSN++
cont., deep VE).

9.89

Inception ↑

Highest IS — beats the best GANs,
no labels.

2.99

bits/dim ↓

Record likelihood via the probability-
flow ODE (sub-VP).

1024^2

first samples

First high-fidelity images from a score-
based model.

One framework — best-in-class samples (VE) and record likelihoods (sub-VP) — from the same score model.

Table 1 — Reverse-Time SDE Solvers (FID)

	Variance Exploding SDE (SMLD)				Variance Preserving SDE (DDPM)			
FID · Sampler	P1000	P2000	C2000	PC1000	P1000	P2000	C2000	PC1000
ancestral sampling	4.98	4.88	—	3.62	3.24	3.24	—	3.21
reverse diffusion	4.79	4.74	20.43	3.60	3.21	3.19	19.06	3.18
probability flow	15.41	10.54	—	3.51	3.59	3.23	—	3.06

Why these numbers: Predictor+Corrector (PC1000) is best everywhere — VP probability-flow PC hits 3.06. Corrector-only (C2000) collapses (≈ 19 – 20) at equal compute: pure MCMC mixes too slowly. Reverse-diffusion beats ancestral sampling, and for VE, predictor-only struggles (15.41) until a corrector rescues it.

Table 2 — Likelihoods (NLL) & FID via the ODE

Model	NLL ↓	FID ↓
RealNVP	3.49	—
Glow	3.35	—
Flow++	3.29	—
Residual Flow	3.28	46.37
DDPM (L_simple)	$\leq 3.75^*$	3.17
DDPM (ours)	3.28	3.37
DDPM cont. (VP)	3.21	3.69
DDPM cont. (sub-VP)	3.05	3.56
DDPM++ cont. (VP)	3.16	3.93
DDPM++ cont. (sub-VP)	3.02	3.16
DDPM++ cont. (deep, VP)	3.13	3.08
DDPM++ cont. (deep, sub-VP)	2.99	2.92

WHY THE LIKELIHOODS IMPROVE

Exact, not ELBO — the ODE gives exact log-likelihood, so the same DDPM beats its own ELBO bound ($3.28 < 3.70$).

Continuous objective helps — training with the continuous loss (DDPM cont.) lowers bits/dim further.

sub-VP wins on likelihood — sub-VP always edges out VP, because its variance never exceeds the VP path.

Record — DDPM++ cont. (deep, sub-VP) sets 2.99 bits/dim — without any maximum-likelihood training.

Table 3 — CIFAR-10 Sample Quality (FID & IS)

Model	FID ↓	IS ↑
Conditional		
BigGAN	14.73	9.22
StyleGAN2-ADA	2.42	10.14
Unconditional		
StyleGAN2-ADA	2.92	9.83
NCSN	25.32	8.87
NCSNv2	10.87	8.40
DDPM	3.17	9.46
DDPM++ cont. (VP)	2.55	9.58
DDPM++ cont. (deep, VP)	2.41	9.68
NCSN++	2.45	9.73
NCSN++ cont. (VE)	2.38	9.83
NCSN++ cont. (deep, VE)	2.20	9.89

WHY VE WINS ON SAMPLES

VE for fidelity — NCSN++ cont. (deep, VE) reaches FID 2.20 / IS 9.89 — the best in the table.

Beats the GANs — it surpasses StyleGAN2-ADA (2.92) unconditionally, and even the best conditional model — with no labels.

Depth + continuous training — deeper NCSN++/DDPM++ and the continuous objective both push FID down.

Mirror image of Table 2 — VE is best for samples while sub-VP is best for likelihood — same score model, different solver/SDE.

APPENDIX

Derivation of the VP Marginals

Why $\alpha(t)$ and $\sigma(t)$ take their closed forms

Linear SDE · Integrating factor · Itô isometry

Appendix — Deriving the Mean $\alpha(t)$

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The VP SDE is linear, so $x(t) \mid x(0)$ is Gaussian. Solve it explicitly — first the mean.

1 · Linear SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}d\mathbf{w}$$

2 · Integrating factor

$$\mu(t) = \exp\left(\frac{1}{2}\int_0^t \beta(s)ds\right) \Rightarrow d(\mu\mathbf{x}) = \mu\sqrt{\beta}d\mathbf{w}$$

3 · Integrate $0 \rightarrow t$

$$\mu(t)\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mu(s)\sqrt{\beta(s)}d\mathbf{w}(s)$$

4 · Solve for $\mathbf{x}(t)$

$$\mathbf{x}(t) = \alpha(t)\mathbf{x}(0) + \alpha(t)\int_0^t \mu(s)\sqrt{\beta(s)}d\mathbf{w}(s), \quad \alpha = \frac{1}{\mu}$$

5 · Take expectation

$$\mathbb{E}[\mathbf{x}(t) \mid \mathbf{x}(0)] = \alpha(t)\mathbf{x}(0), \quad \alpha(t) = \exp\left(-\frac{1}{2}\int_0^t \beta(s)ds\right)$$

The Itô integral has zero mean, so the conditional mean is simply $\alpha(t)\cdot x(0)$. Continues on the next slide →

Appendix — Deriving the Variance $\sigma^2(t)$

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The variance of the same solution, from the stochastic integral via Itô isometry.

1 · Random part

$$\mathbf{x}(t) - \alpha(t)\mathbf{x}(0) = \alpha(t) \int_0^t \mu(s) \sqrt{\beta(s)} d\mathbf{w}(s)$$

2 · Itô isometry

$$\text{Var}[\mathbf{x}(t)] = \alpha^2(t) \int_0^t \mu^2(s) \beta(s) ds = \alpha^2(t) \int_0^t e^{\int_0^s \beta(u) du} \beta(s) ds$$

3 · Perfect derivative

$$e^{\int_0^s \beta} \beta(s) = \frac{d}{ds} e^{\int_0^s \beta} \Rightarrow \int_0^t (\cdot) ds = e^{\int_0^t \beta} - 1 = \alpha^{-2}(t) - 1$$

4 · Collect

$$\sigma^2(t) = \alpha^2(t) (\alpha^{-2}(t) - 1) = 1 - \alpha^2(t)$$

Variance preserving: with normalized data, $\alpha^2(t) + \sigma^2(t) = 1$, so the marginal variance of $\mathbf{x}(t)$ stays ≈ 1 throughout the process — the property that names the VP SDE.

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